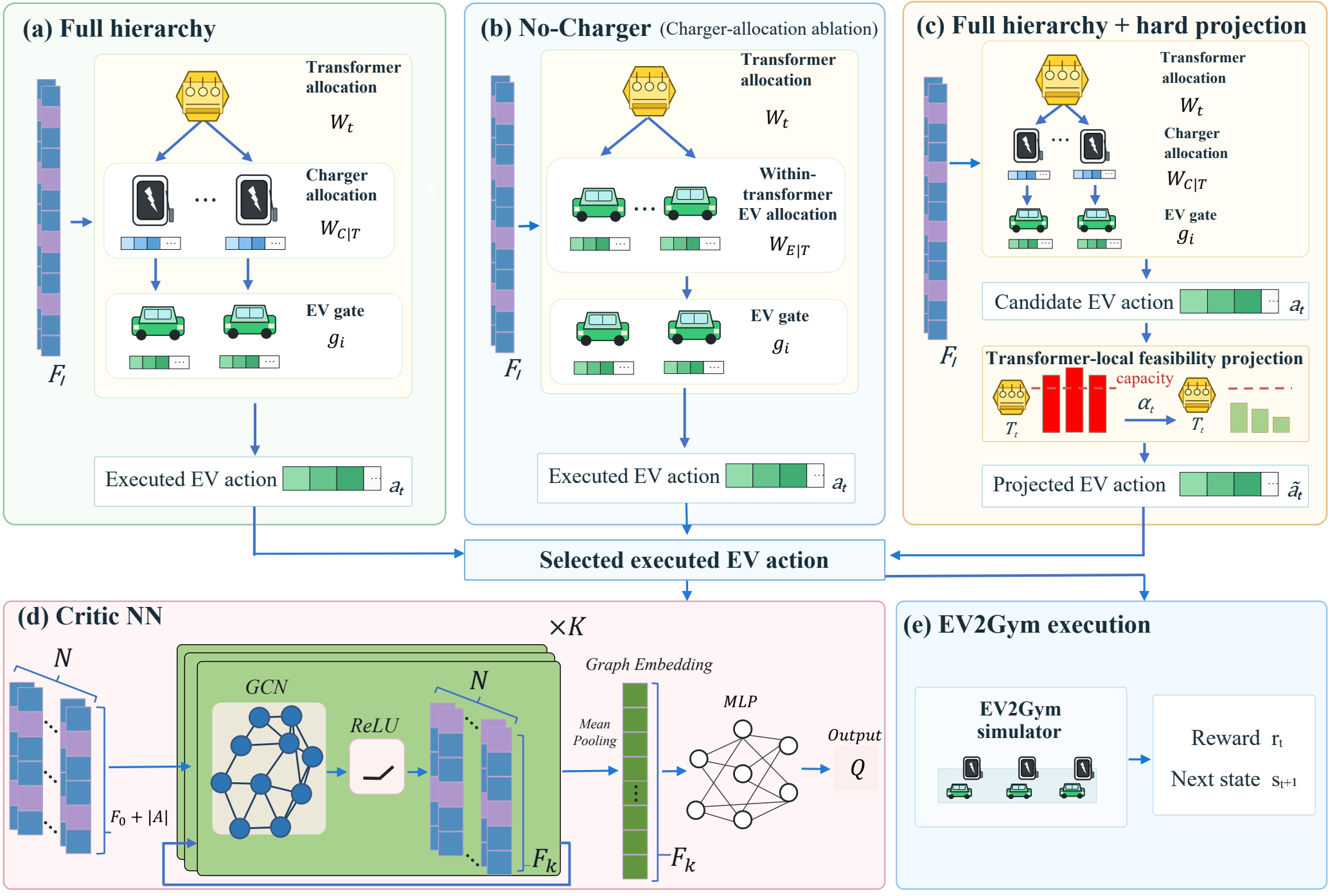

(a) Full hierarchy
Transformer allocation
W_t
Charger allocation
$W_{C|T}$
EV gate
g_i
F_l
Executed EV action
a_t
(b) No-Charger (Charger-allocation ablation)
Transformer allocation
W_t
Within-transformer EV allocation
$W_{E|T}$
EV gate
g_i
F_l
Executed EV action
a_t
(c) Full hierarchy + hard projection
Transformer allocation
W_t
Charger allocation
$W_{C|T}$
EV gate
g_i
F_l
Candidate EV action
a_t
Transformer-local feasibility projection
capacity
α_t
T_t
T_t
Projected EV action
$\tilde{a}_t$
Selected executed EV action
(d) Critic NN
$\times K$
N
$F_0 + |A|$
GCN
ReLU
N
F_k
Graph Embedding
Mean Pooling
MLP
Output
Q
F_k
(e) EV2Gym execution
EV2Gym simulator
Reward r_t
Next state s_{t+1}